

FORMS OF ENERGY

Stored Energy and Energy in Transition

- Two categories of energy
- Stored energy is the energy contained by the system
(P.E, K.E, I.E , electrical energy, etc)
- Energy in transition is the energy that is getting transferred to or from the system (Heat and Work)

Forms of Energy



- Energy can exist in numerous forms such as thermal, mechanical, kinetic, potential, electric, magnetic, chemical, and nuclear, and their sum constitutes the **total energy E of a system**
- **Total energy, $E = \text{Macroscopic Energy} + \text{Microscopic Energy}$**
- Thermodynamics provides no information about the absolute value of the total energy. It deals only with the change of the total energy

- The macroscopic forms of energy are those a system possesses as a whole with respect to some outside reference frame, such as kinetic and potential energies
- The microscopic forms of energy are those related to the molecular structure of a system and the degree of the molecular activity, and they are independent of outside reference frames. The sum of all the microscopic forms of energy is called the **Internal energy** of a system and is denoted by **U**

$$E = U + \text{KE} + \text{PE} = U + m \frac{V^2}{2} + mgz \quad (\text{kJ})$$

- Energy change = Energy at final state - Energy at initial state

$$\Delta E_{\text{system}} = E_{\text{final}} - E_{\text{initial}} = E_2 - E_1$$

Internal Energy (U)

- Is the sum of all microscopic form of energy with in the system (degree of molecular activity, since it is stored in the molecular or atomic structure of the system).
- It is mainly , the sum of potential and kinetic energy of the molecules of the system.
- In fact a substance does not contain heat – it contains internal energy
- It is a point function and hence a property of the system
- It is expressed in J or kJ, and internal energy per unit mass is called specific internal energy

ENTHALPY (H)

- Measure of the total energy of a thermodynamic system
- Sum of the internal energy and the product pressure and volume
- $H = U + pV$
- Enthalpy is also a property of the system
- Unit is kJ
- Enthalpy per unit mass is called specific enthalpy(h)
- $h = u + pv$

HEAT and WORK

- Energy can cross the boundary of a system in two ways:
- (1) Work
- (2) Heat

HEAT

- Heat is defined as the form of energy that is transferred between two systems (or a system and its surroundings) by virtue of a temperature difference
- The transfer of heat into a system is frequently referred to as heat addition and the transfer of heat out of a system as heat rejection
- It is a transient phenomenon
- Heat is transferred by three mechanisms: Conduction, Convection, and Radiation
- A process during which there is no heat transfer is called an adiabatic process
- It is measured in Joules(J) or kJ

Heat and Internal Energy

- Two categories of energy
- Internal energy is a property whereas heat is not
- A body may contain energy(in stored form), but not heat .
- Internal energy is a property of the system (point function) and its value depends only on the state of the system.
- Heat is a path function and its value depends on the path of the process in which the heat transfer is taking place.

SPECIFIC HEATS

- The specific heat is defined as the heat energy required to change the temperature of a unit mass of a substance by one degree .
- Its unit is kJ/kgK
- The specific heat at constant volume, C_v is defined as the energy required to raise the temperature of the unit mass of a substance by one degree as the volume is maintained constant
- The specific heat at constant pressure, C_p is defined as the energy required to raise the temperature of the unit mass of a substance by one degree as the pressure is maintained constant

WORK

- Work, like heat, is an energy interaction between a system and its surroundings
- The energy transfer across the boundary of the system, other than due to temperature difference , is called work.
- Positive work is done by a system when the sole effect external to the system can be reduced to the raising of a weight
- Work changes the state of a system and is a path function, hence it is not a thermodynamic property

WORK

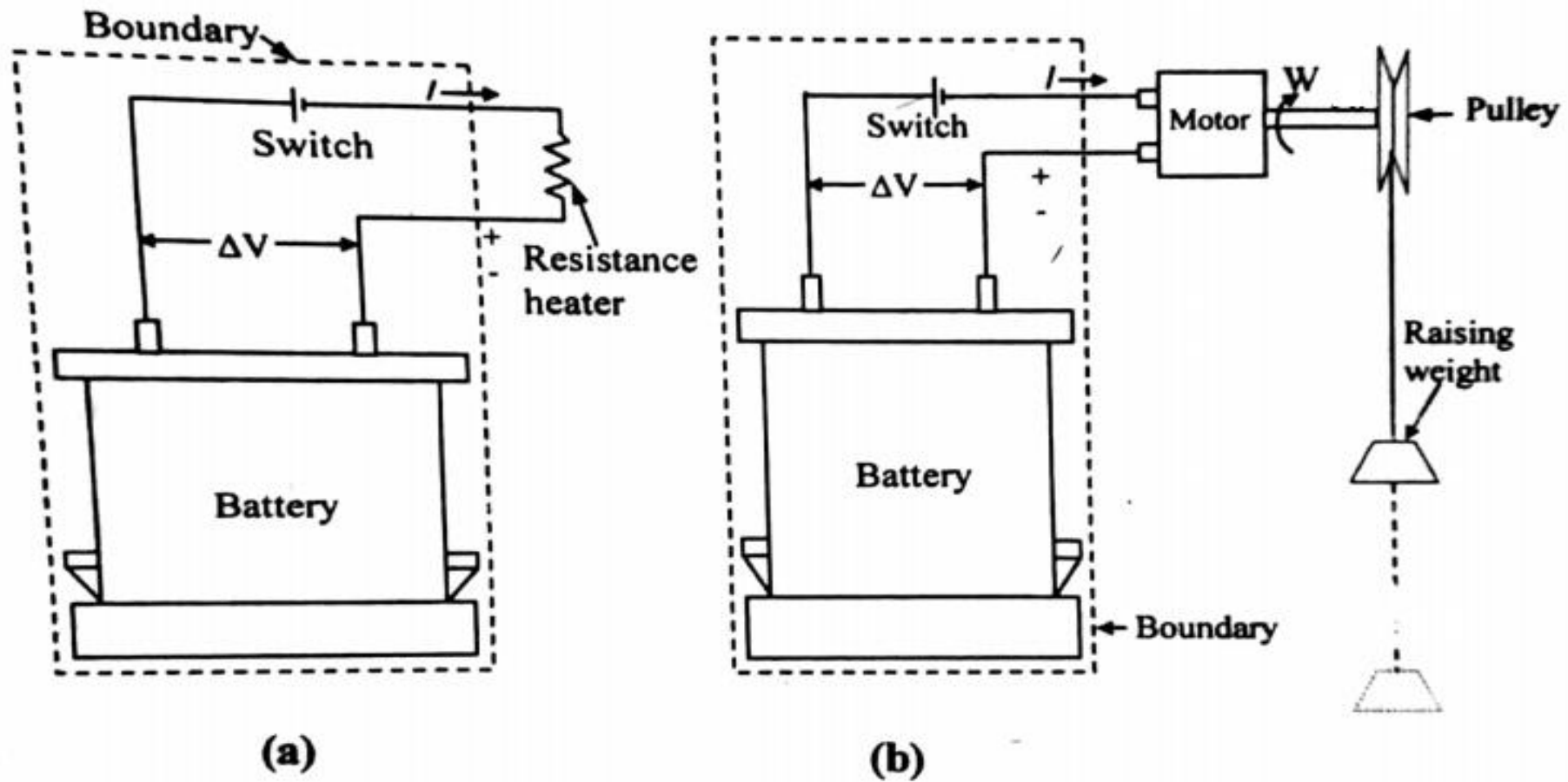


Fig. *Illustration of thermodynamic work*

WORK

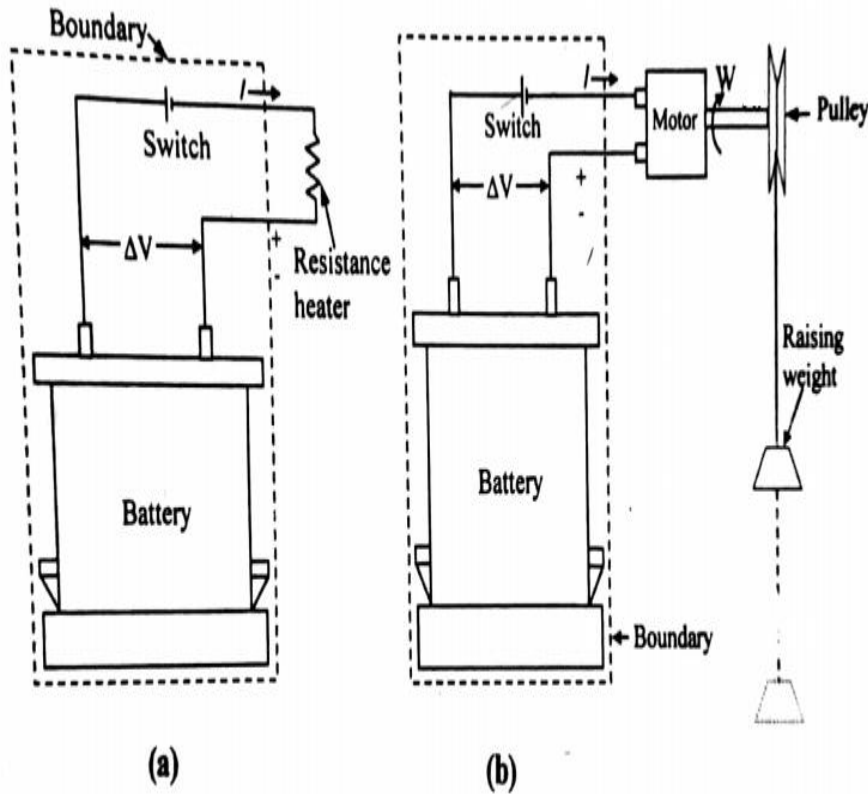


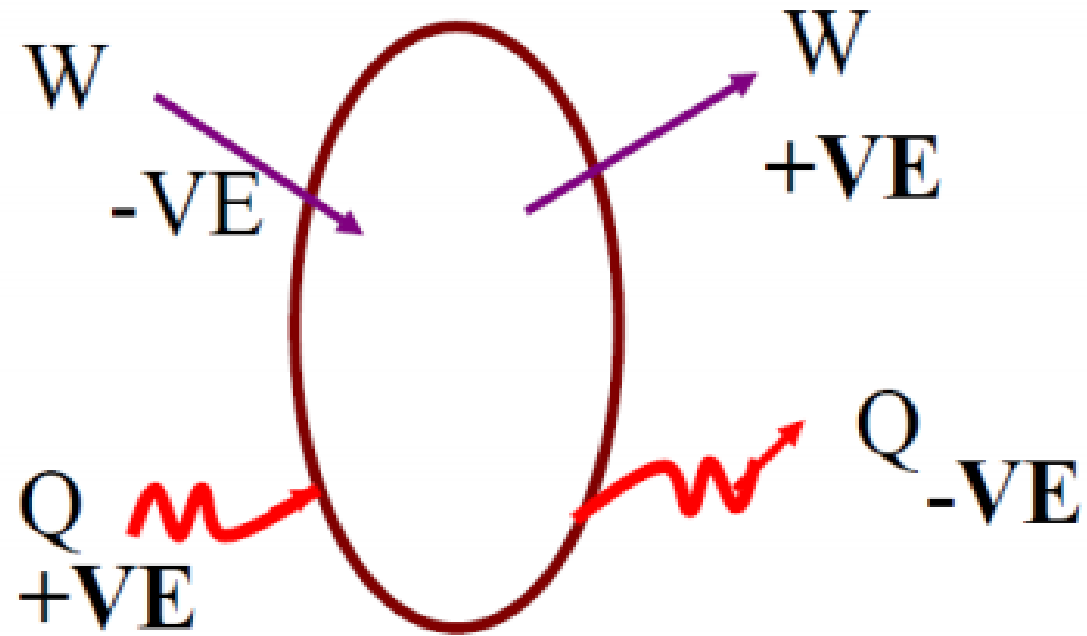
Fig. Illustration of thermodynamic work

- Consider a system consisting of an electrical battery, switch and a resistance coil outside the system
- When the switch is closed, current flows to the resistance and is converted to heat, **ie work is done by the system.**
- This can be illustrated by replacing the external resistance by an imaginary motor, pulley and weight arrangement
- When the switch is closed, the motor and pulley rotates and raises the suspended weight.
- Thus **the sole effect external to the system becomes raising of a weight ie the system does work**

Similarities Between Heat and Work

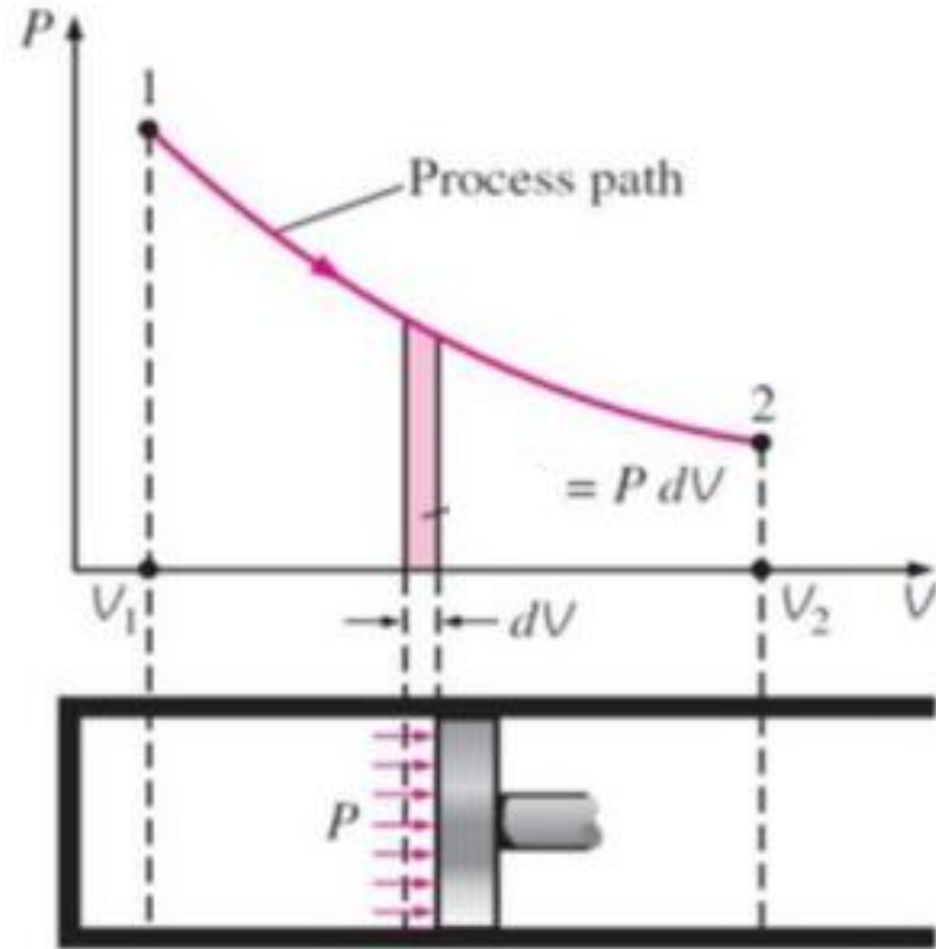
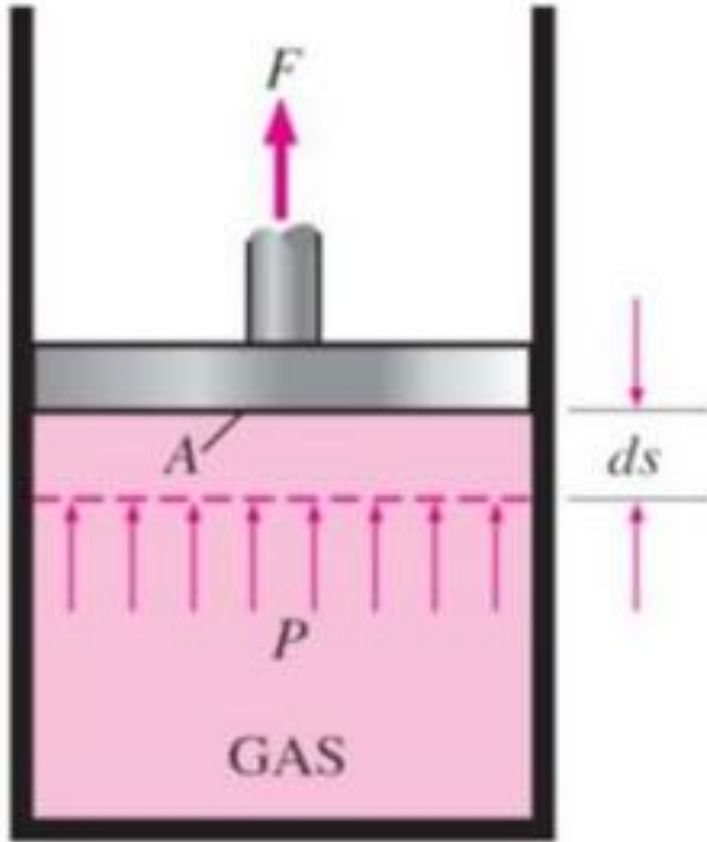
- Both are recognized at the boundaries of a system as they cross the boundaries. That is, both heat and work are boundary phenomena
- Systems possess energy, but not heat or work
- Both are associated with a process, not a state. Unlike properties, heat or work has no meaning at a state
- Both are path functions (i.e., their magnitudes depend on the path followed during a process as well as the end states)
- Heat is low grade energy and work is high grade energy

Sign Conventions



- Heat transfer to a system and work done by a system are **positive**; heat transfer from a system and work done on a system are **negative**

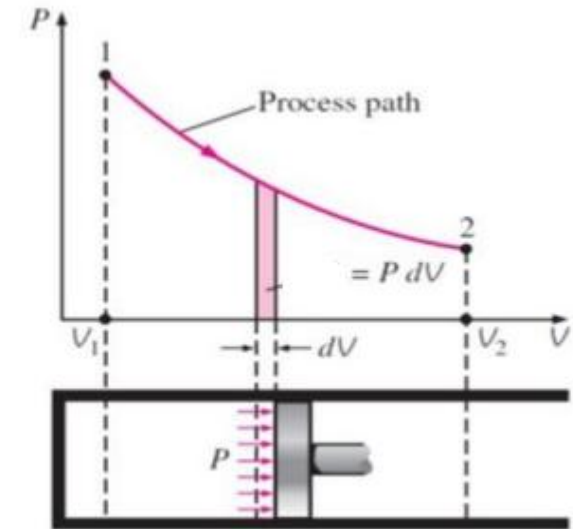
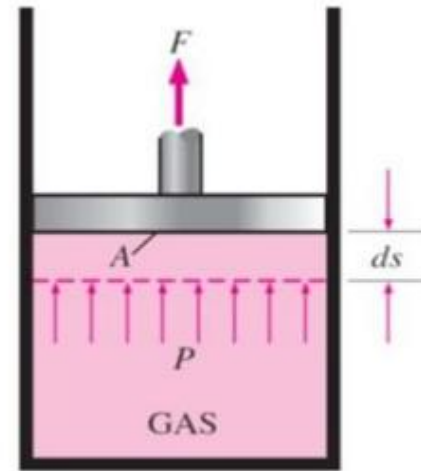
Displacement Work ($p dV$ work)



- Consider a gas enclosed in a piston- cylinder arrangement. The gas pressure acts on the piston and moves it.
- Force applied on the piston , $F = pA$
- Differential work transfer through a piston displacement of ' ds ' during the process,

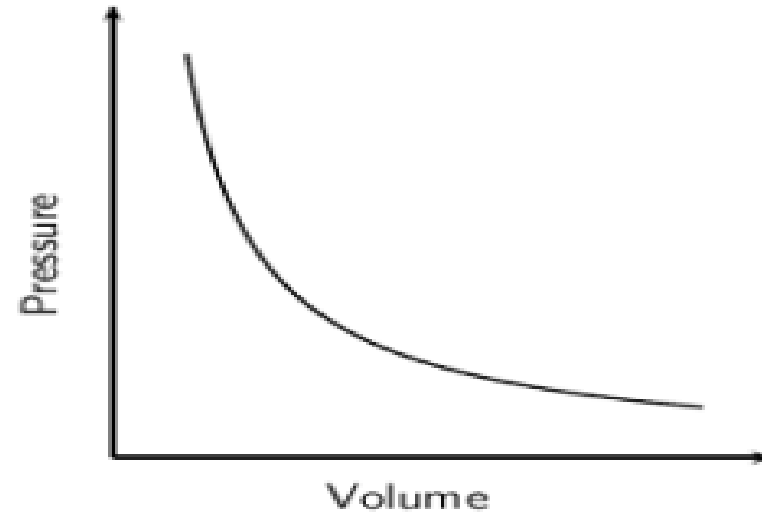
$$\delta W = F \times ds = pA \times ds = pdv \quad (A \times ds = dv, \text{ change in volume })$$

- For a closed system, Work, δW or $W_{1-2} = \int_1^2 P dV$, area under the curve 1-2



Gas laws

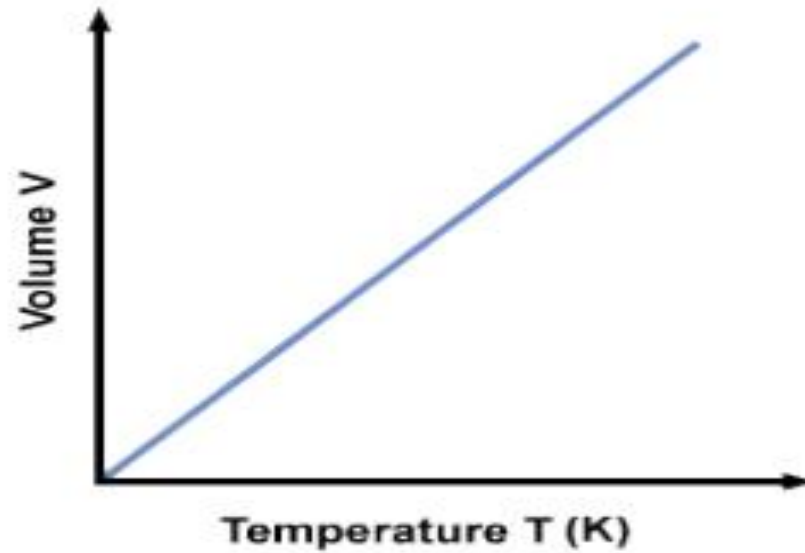
Boyle's Law



- Formulated by Robert Boyle in 1662
- *It states that absolute pressure of a given mass of a perfect gas varies inversely as its volume, when the temperature remains constant*
- **$V \propto 1/P$, $T = \text{constant}$**
- **$PV = \text{constant}$, $T = \text{constant}$**
- **$P_1V_1 = P_2V_2 = P_3V_3 = PV = \text{constant}$, $T = \text{constant}$**

Gas laws

Charle's Law



- Formulated by A.C Charles in 1787
- *It states that, the volume of a given mass of a perfect gas varies directly as its absolute temperature, if the pressure remains constant*
- **$V \propto T$, $P = \text{constant}$**
- **$V/T = \text{constant}$, $P = \text{constant}$**
- **$V_1/T_1 = V_2/T_2 = V_3/T_3 = V/T = \text{constant}$, $P = \text{constant}$**

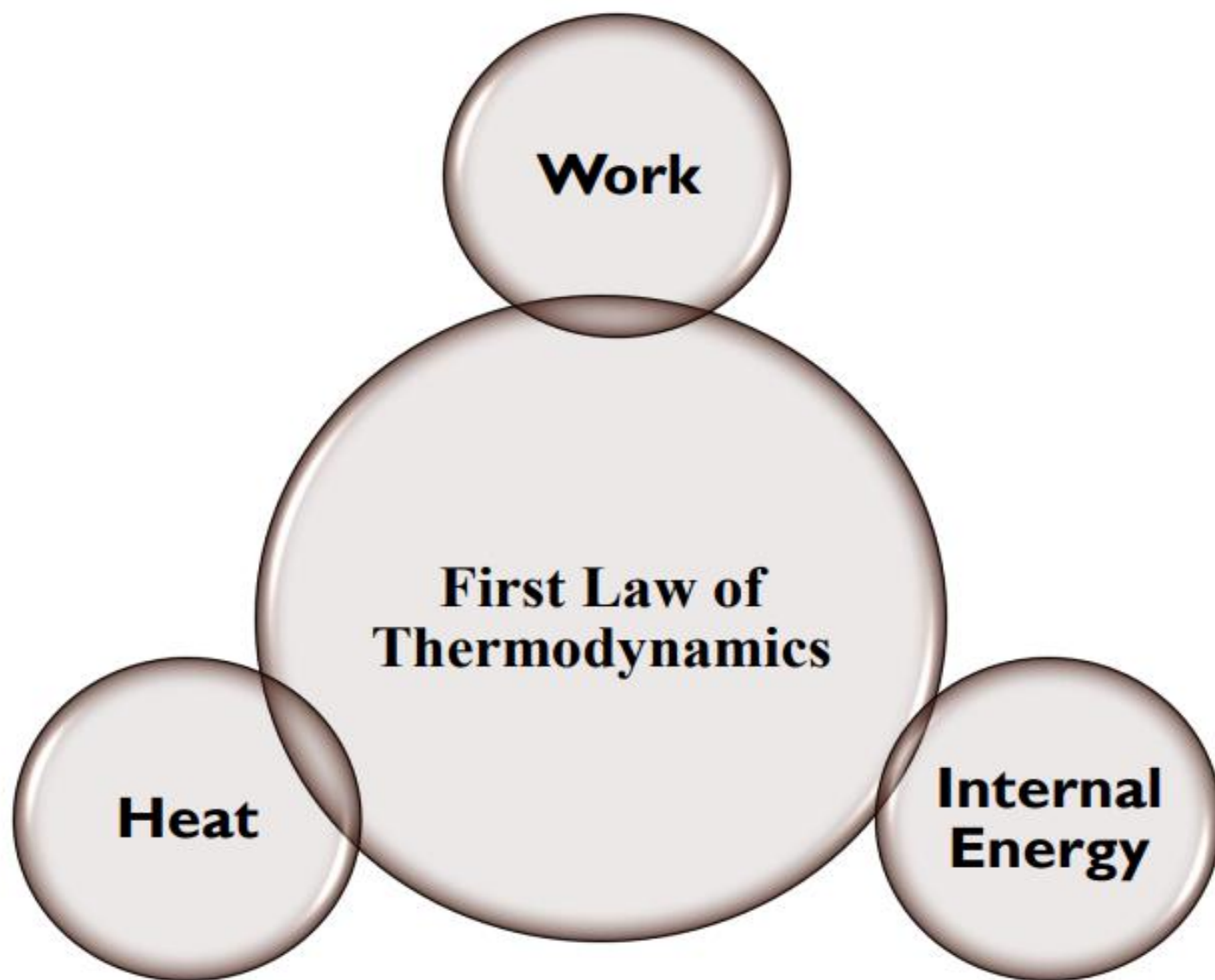
Characteristic Gas Equation

- $PV = mRT$ (Characteristic gas equation)
; m-mass, R-characteristic gas constant

For air $R = 287 \text{ J/KgK}$

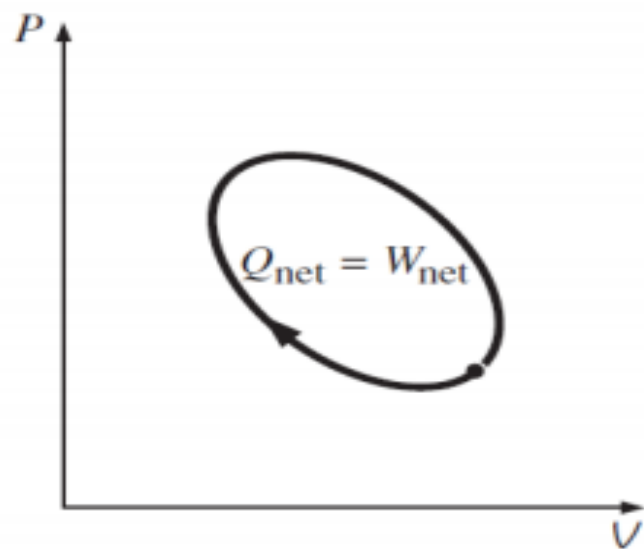
Universal Gas Constant

- Product of molecular weight and characteristic gas constant of any gas is constant
- $R_u = R \times M$
- $R_u = 8314 \text{ J/Kg mol K}$



The First Law of Thermodynamics

- The first law of thermodynamics, also known as the conservation of energy principle, provides a sound basis for studying the relationships among the various forms of energy and energy interactions
- *First law states that, When a closed system executes a complete cycle **the sum of heat interactions is equal to the sum of work interactions***
- $\Sigma Q = \Sigma W$; $Q_{\text{net}} = W_{\text{net}}$; $\oint \delta Q = \oint \delta W$



First law applied to a process

- If a system undergoes a change of state during which both heat transfer and work transfer are involved, then the net energy transferred will be stored or accumulated within the system
- If Q is the amount of heat transferred to the system and W is the work transferred from the system during a process, the net energy ($Q-W$) will get stored in the system.

$$Q-W = \Delta E$$

ie, $Q - W = \Delta U.$ ($\Delta E = \Delta U$)

- When heat is added into a system it can either 1) change the internal energy of the system (i.e. make it hotter) or 2) go into doing work.

$$Q = W + \Delta U$$

- $\delta Q - \delta W = dU$ or $Q_{1-2} = W_{1-2} + dU$
- For a constant volume process from 1 to 2, $W_{1-2} = 0$
- $Q_{1-2} = dU$; $Q_{1-2} = mc_v(T_2 - T_1)$
- So, $dU = mc_v(T_2 - T_1)$
- Change of internal energy is directly proportional to change of temperature. For an ideal gas the internal energy is a function of the temperature only
- Internal energy is a point function. Increase of internal energy of the gas is considered as positive and decrease of internal energy of the gas is considered as negative

SPECIFIC HEAT OF IDEAL GAS

Specific heat at Constant Volume (C_v)

Specific heat at constant volume of a gas is defined as the heat energy transferred to change the temperature of unit mass of the gas by one degree, when the volume of the gas is maintained constant.

or
$$C_v = \left(\frac{\delta Q}{dT} \right)_v$$

where δQ is the heat transferred and dT is the corresponding change in temperature.

Consider unit mass of a gas undergoing change of state at constant volume. From the first law applied to a process,

$$\begin{aligned}\delta Q &= \delta W + du \\ &= p dv + du\end{aligned}$$

Since the process is at constant volume, $dv = 0$. Therefore,

$$\delta Q = du$$

Hence, the definition of C_v ,

$$C_v = \left(\frac{du}{dT} \right)$$

i.e. C_v can be defined as the change of specific internal energy of a substance per unit change in temperature. Also, we can write the change in internal energy of a gas as,

$$du = C_v dT$$

Specific heat at Constant Pressure (C_p)

Specific heat at constant pressure of a gas is defined as the amount of heat energy transferred to change the temperature of unit mass of the gas by one degree when the pressure is maintained constant during the process.

or
$$C_p = \left(\frac{\delta Q}{dT} \right)_p$$

where δQ is the heat transferred and dT is the corresponding change in temperature.

Consider a gas undergoing change of state at constant pressure. From the definition of enthalpy

$$h = u + pv$$

In differential form, it is

$$dh = du + p dv + v dp$$

$$= du + p dv$$

($\because dp = 0$ for the constant pressure process)

$$= \delta Q$$

(From first law for a process)

Hence, the definition of C_p may be written as,

$$C_p = \left(\frac{dh}{dT} \right)$$

i.e., C_p can be defined as the change in the specific enthalpy of a substance per unit change in temperature. Also, we can write the change in specific enthalpy of a gas as,

$$dh = C_p dT$$

Relation Between Specific heats for Ideal Gas

From the definition of enthalpy,

$$h = u + pv$$

or

$$h = u + RT$$

(For an ideal gas, $pv = RT$)

In the differential form, the equation is,

$$dh = du + R dT$$

(R is a constant for any gas)

Replacing dh by $C_p dT$ and du by $C_v dT$, we get

$$\begin{aligned} C_p dT &= C_v dT + R dT \\ &= (C_v + R) dT \end{aligned}$$

or

$$C_p - C_v = R$$

Ratio of Specific Heats (Adiabatic Index)

The **Adiabatic index** of a gas is the ratio of its specific heat at constant pressure to specific heat at constant volume. It is denoted by greek letter gamma (γ).

$$\gamma = C_p / C_v$$

Since C_p is greater than C_v , γ for all gases are greater than unity.

we have,

$$C_p - C_v = R$$

$$C_v \left(\frac{C_p}{C_v} - 1 \right) = R$$

$$C_v (\gamma - 1) = R$$

$$C_v = R / (\gamma - 1)$$

or

Also,

$$C_p \left(1 - \frac{C_v}{C_p} \right) = R$$

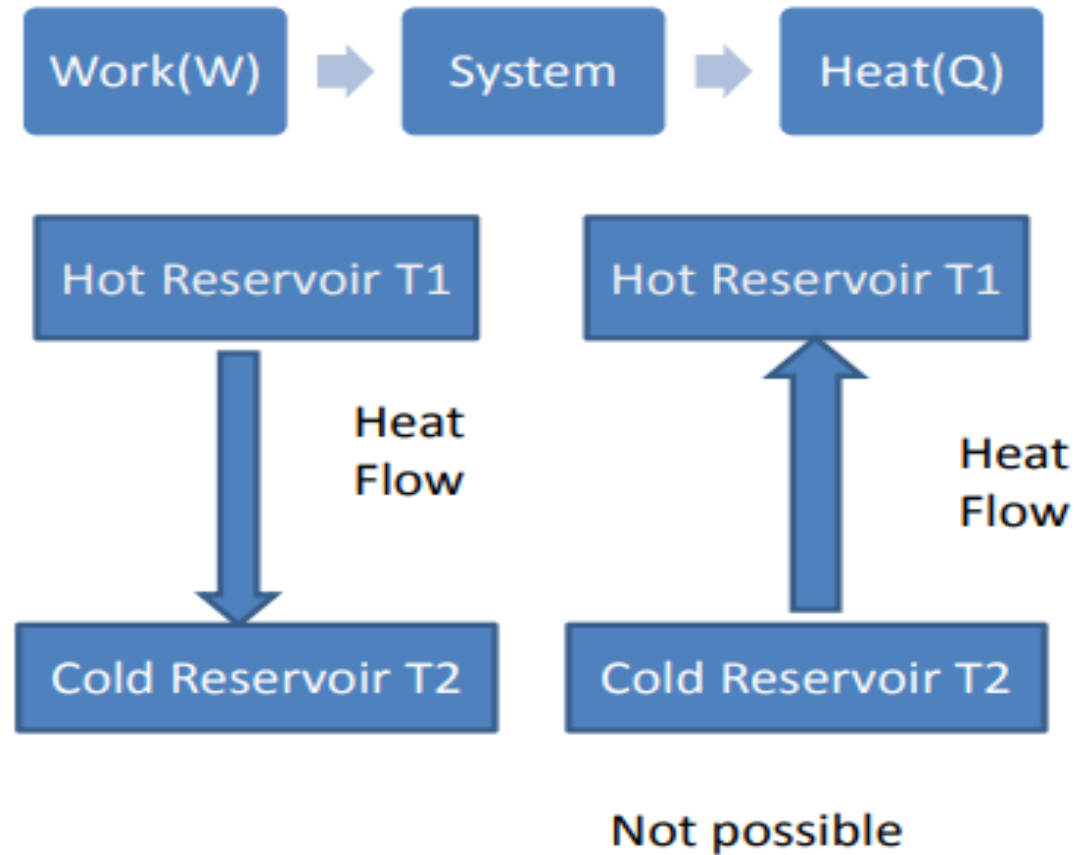
$$C_p \left(1 - \frac{1}{\gamma} \right) = R$$

$$C_p \left(\frac{\gamma - 1}{\gamma} \right) = R$$

$$C_p = \frac{\gamma}{\gamma - 1} R$$

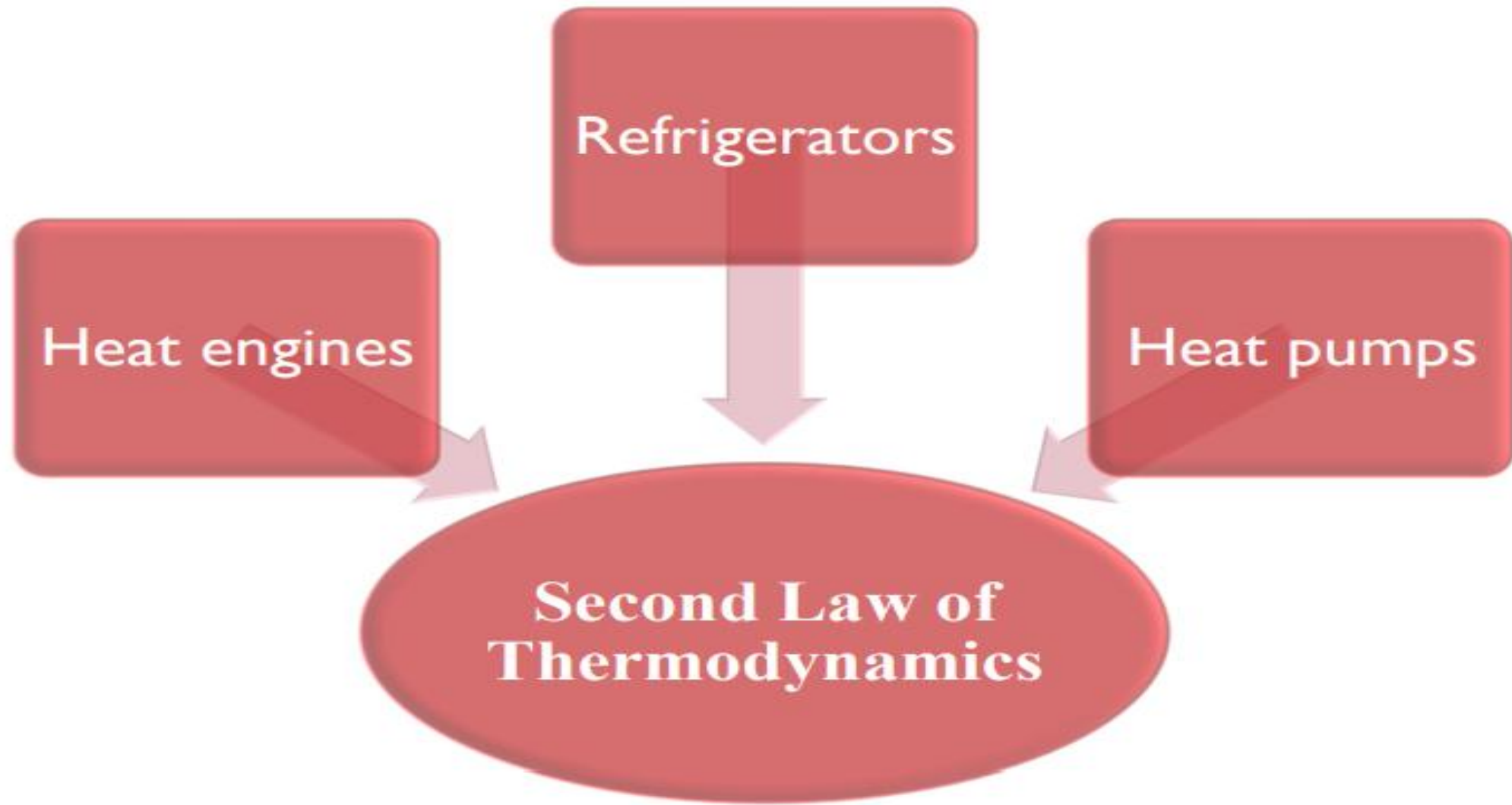
Limitation of “FIRST LAW”

- First law does not help to predict whether the certain process is possible or not.
- The first law does not give info about Direction.
- It does not provide and specify sufficient condition to process take place.



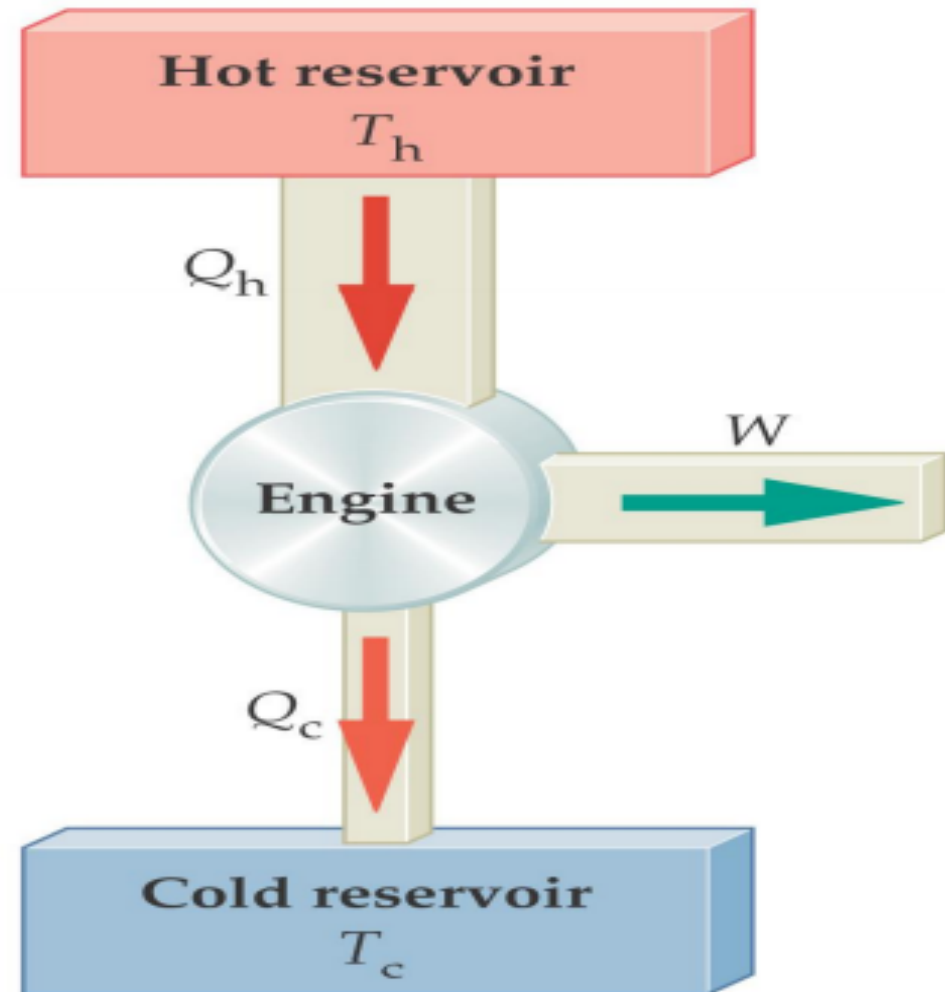
LIMITATIONS OF FIRST LAW OF THERMODYNAMICS

- The first law of thermodynamics is a law of conservation of energy only. It does not specify any direction for heat flow. *But it is known that heat , by itself, can't flow from low temperature systems to higher temperature surroundings*
- First law of thermodynamics does not stipulate any restriction in the efficiency of heat to work conversion.
- there is a limitation on the conversion of one form of energy in to other , about which the first law is silent.
- ***The second law takes in to account the direction of process as well as the quality of energy***



Introduction to the Second Law of Thermodynamics

- A process cannot occur unless it satisfies both the first and the second laws of thermodynamics
- The second law also asserts that energy has **quality** as well as **quantity**

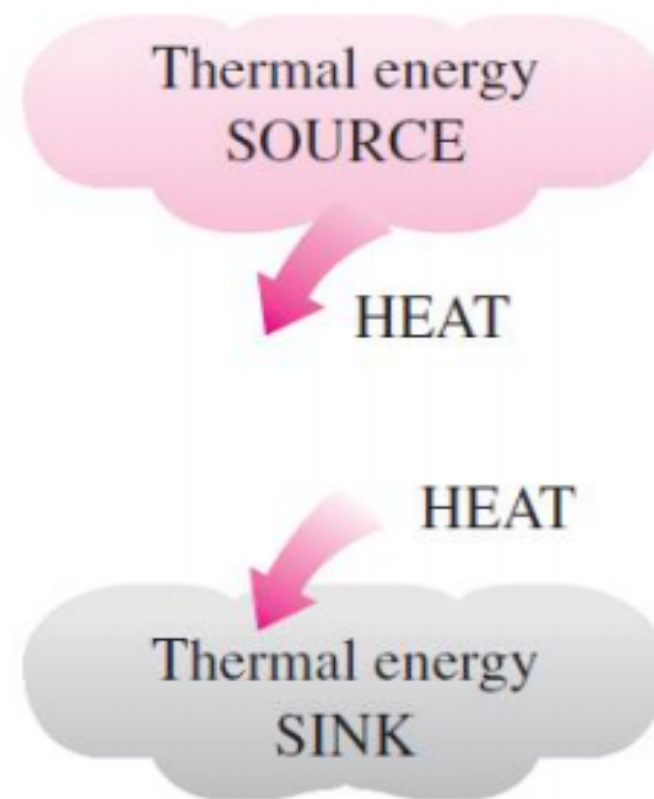


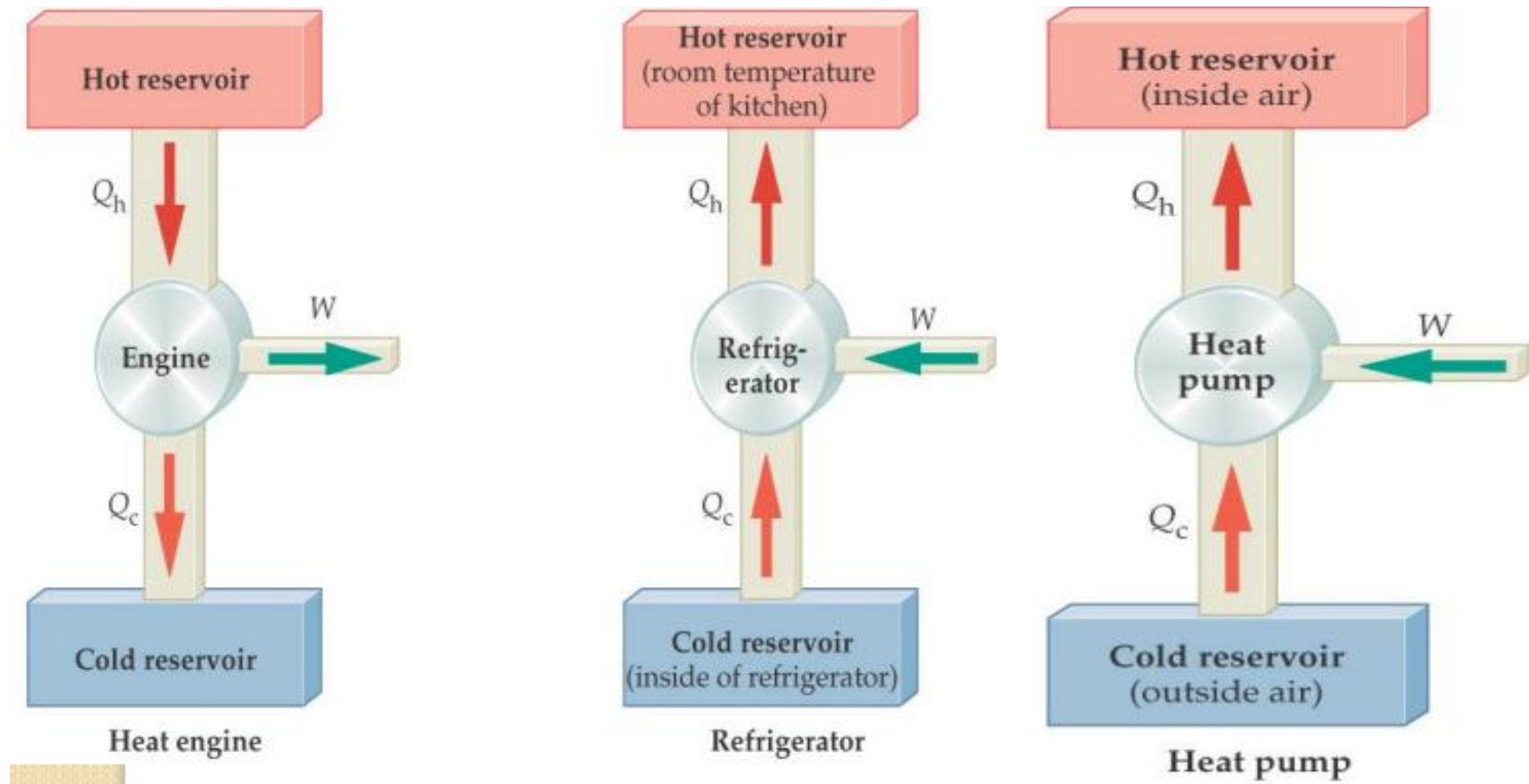
The “**SECOND LAW**” of thermodynamic:

- The second law of thermodynamic gives more information about thermodynamic processes.
- Second law may be defined as
 - “***Heat can not flow itself from colder body to a hotter body***”.
- The Second law is also used to determine the theoretical limits for the performance of mostly used engineering systems like heat engines and heat pump....

Thermal Reservoirs

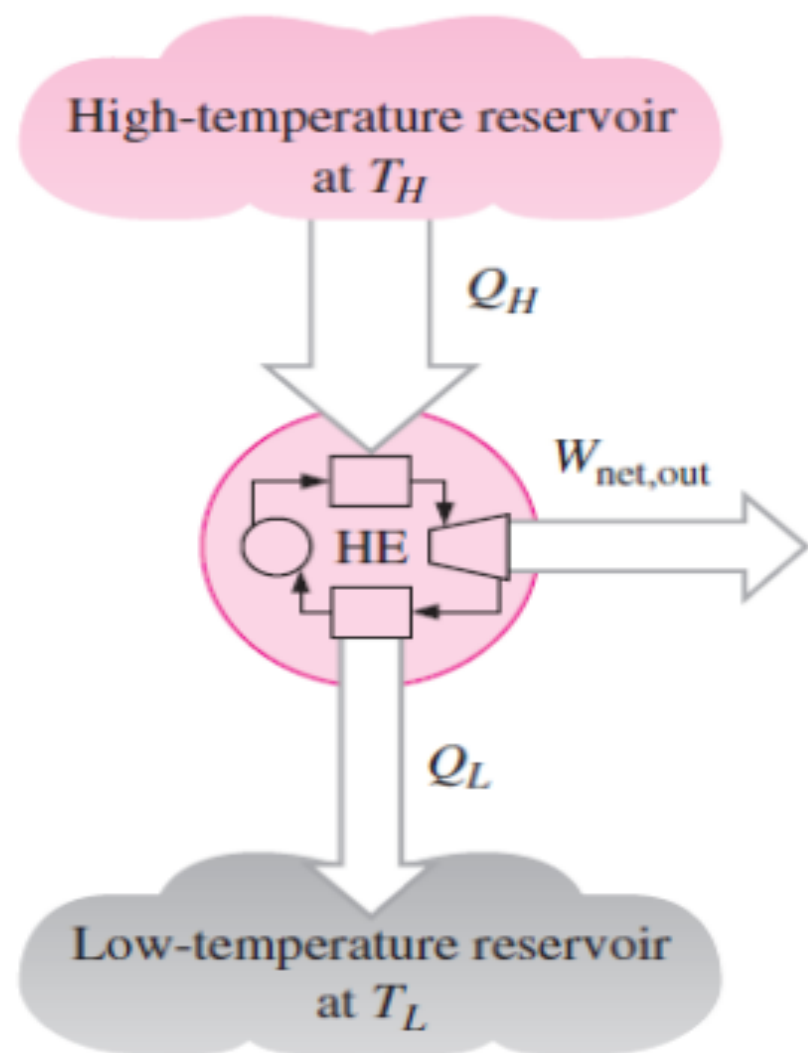
- In the development of the second law of thermodynamics, it is very convenient to have a hypothetical body with a relatively large thermal energy capacity (mass $\times$ specific heat) that can supply or absorb finite amounts of heat without undergoing any change in temperature. Such a body is called a **thermal energy reservoir**, or just a reservoir. Eg. Atmosphere, furnace...etc
- A reservoir that supplies energy in the form of heat is called a **source**, and one that absorbs energy in the form of heat is called a **sink**





Heat Engines

- They receive heat from a high-temperature source
- They convert part of this heat to work
- They reject the remaining waste heat to a low-temperature sink (the atmosphere, rivers, etc.).
- They operate on a cycle.
- Heat engines and other cyclic devices usually involve a fluid to and from which heat is transferred while undergoing a cycle. This fluid is called the **working fluid**
- Eg. Internal combustion engine, gas turbine



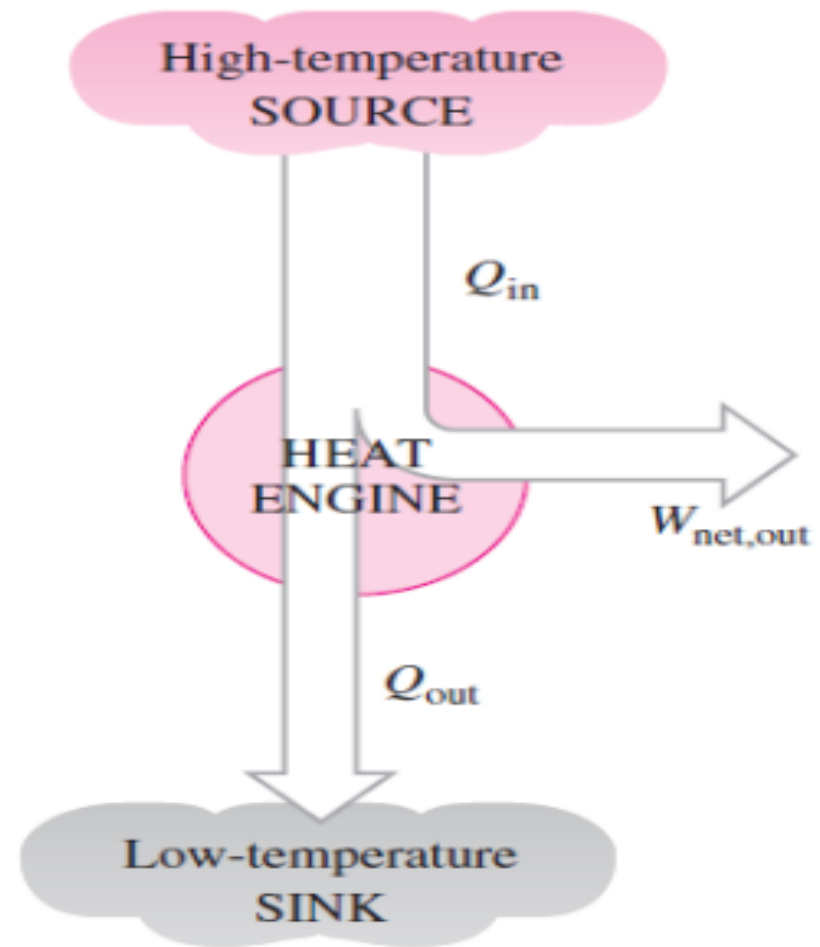
Thermal Efficiency

- The fraction of the heat input that is converted to net work output is a measure of the performance of a heat engine and is called the **thermal efficiency (η_{th})**

$$\text{Thermal efficiency} = \frac{\text{Net work output}}{\text{Total heat input}}$$

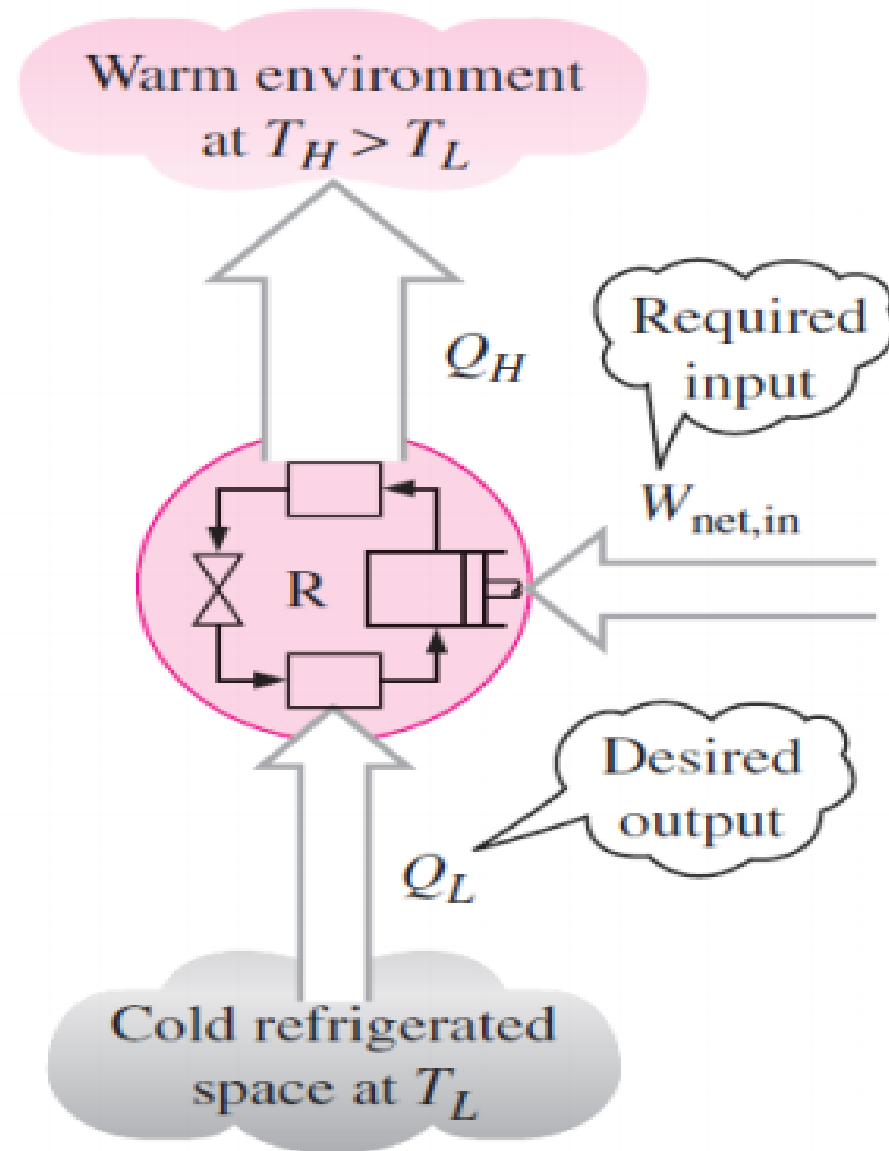
$$\eta_{th} = \frac{W_{net,out}}{Q_{in}}$$

$$\eta_{th} = 1 - \frac{Q_{out}}{Q_{in}}$$



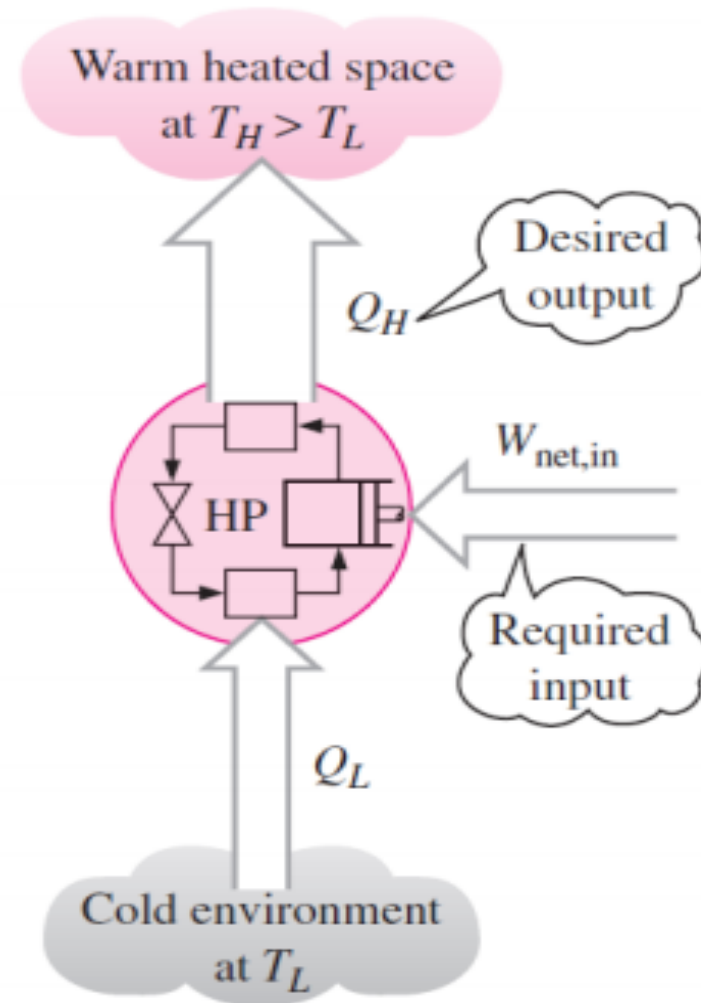
Refrigerators

- The transfer of heat from a low-temperature medium to a high-temperature one requires special devices called **refrigerators**
- The objective of a refrigerator is to maintain the refrigerated space at a low temperature by removing heat from it
- Refrigerators, like heat engines, are cyclic devices. The working fluid used in the refrigeration cycle is called a **refrigerant**



Heat Pumps

- Another device that transfers heat from a low-temperature medium to a high-temperature one is the **heat pump**
- The objective of a heat pump, however, is to maintain a heated space at a high temperature. This is accomplished by absorbing heat from a low-temperature source, such as well water or cold outside air in winter, and supplying this heat to the high-temperature medium such as a house



STATEMENTS OF SECOND LAW

- Clausius Statement

It is impossible for a self acting machine , operating in a cycle, to supply heat from a body at a lower temperature to a body at a higher temperature, without the aid of an external agency

- Kelvin- Plank Statement

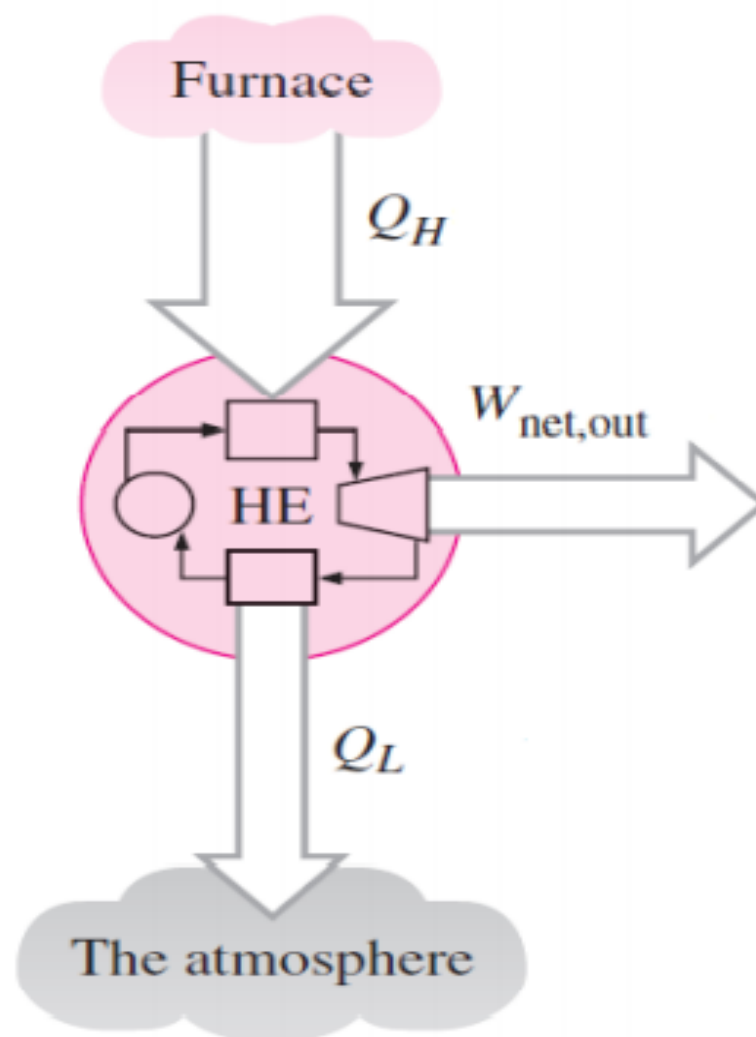
It is impossible to construct an engine working on a cyclic process whose sole purpose is to convert all the heat supplied to it in to equivalent amount of work

The Second Law of Thermodynamics

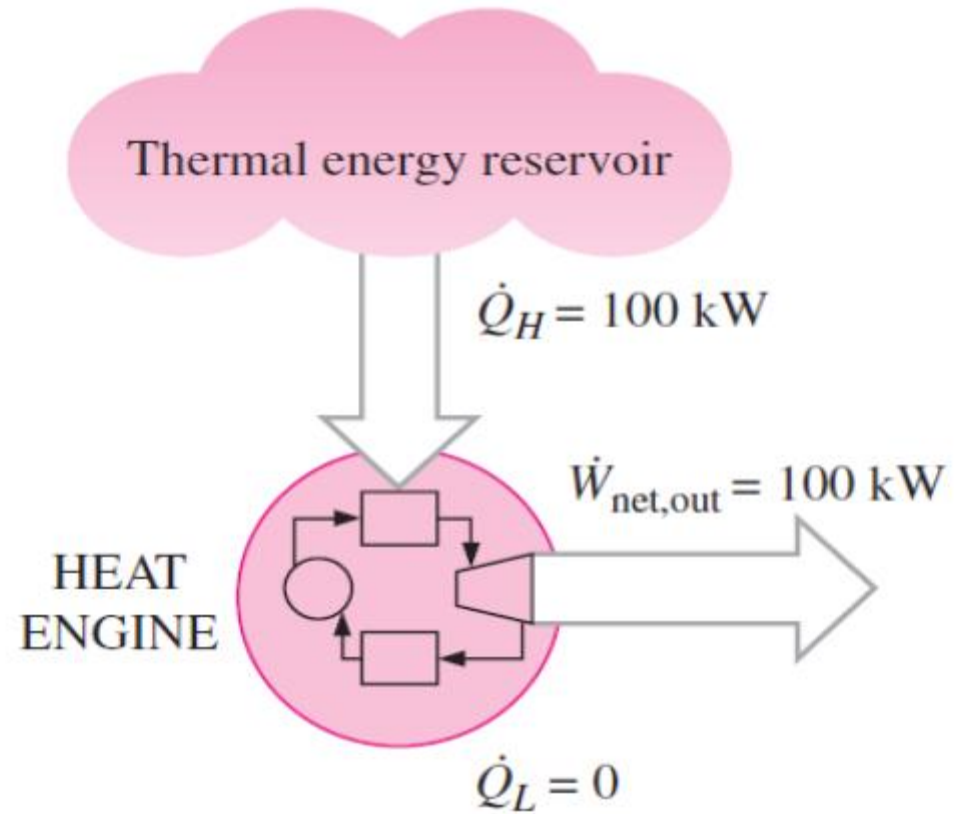
1. Kelvin–Planck Statement

It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work

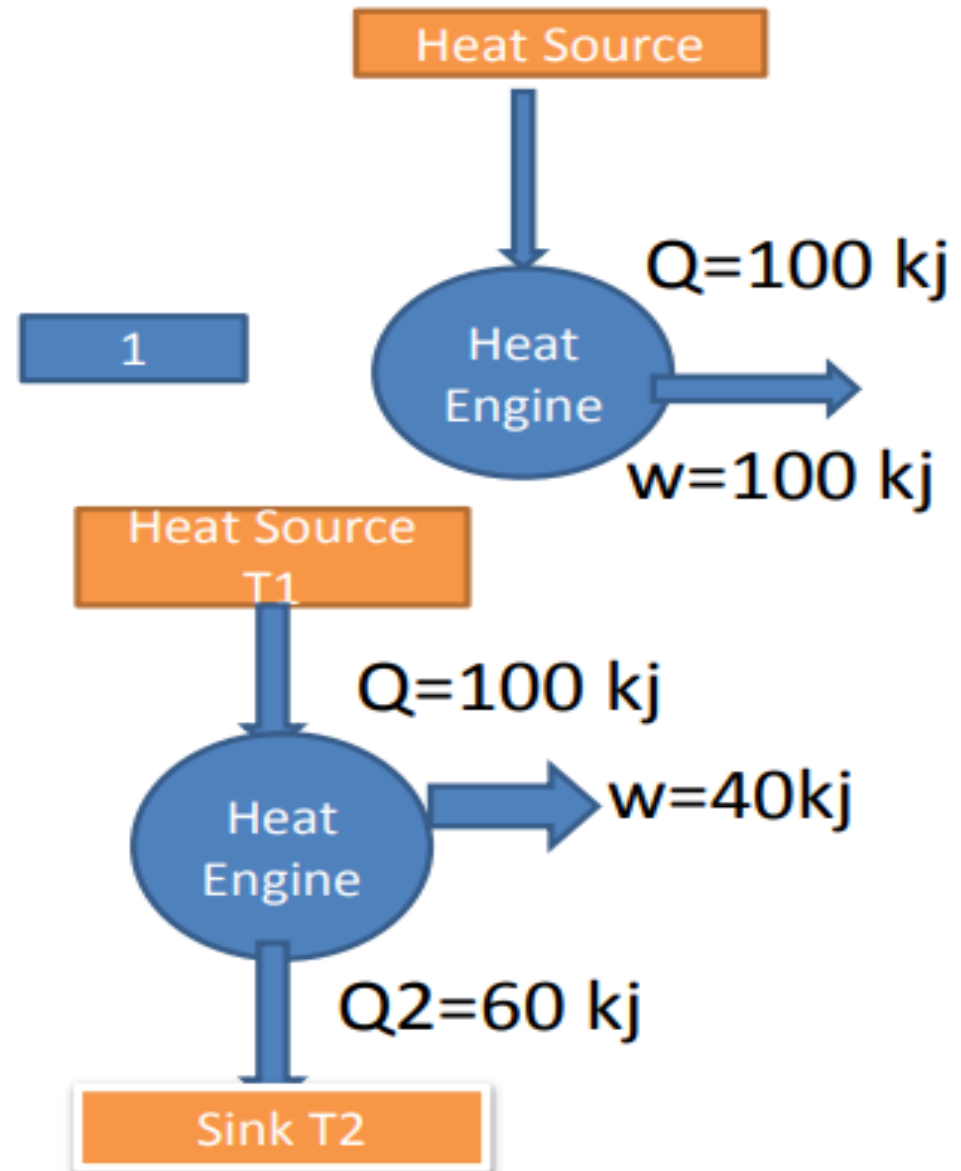
- That is, a heat engine must exchange heat with a low-temperature sink as well as a high-temperature source to keep operating. The Kelvin–Planck statement can also be expressed as **no heat engine can have a thermal efficiency of 100 percent**



Violation of Kelvin - Planck statement



- The heat engine as shown in fig.1 is converting 100 % of heat into 100% work.
- This system is not satisfying second law.
- So, this statement can be also expressed as “**No heat engine has a 100 % of thermal efficiency**”.

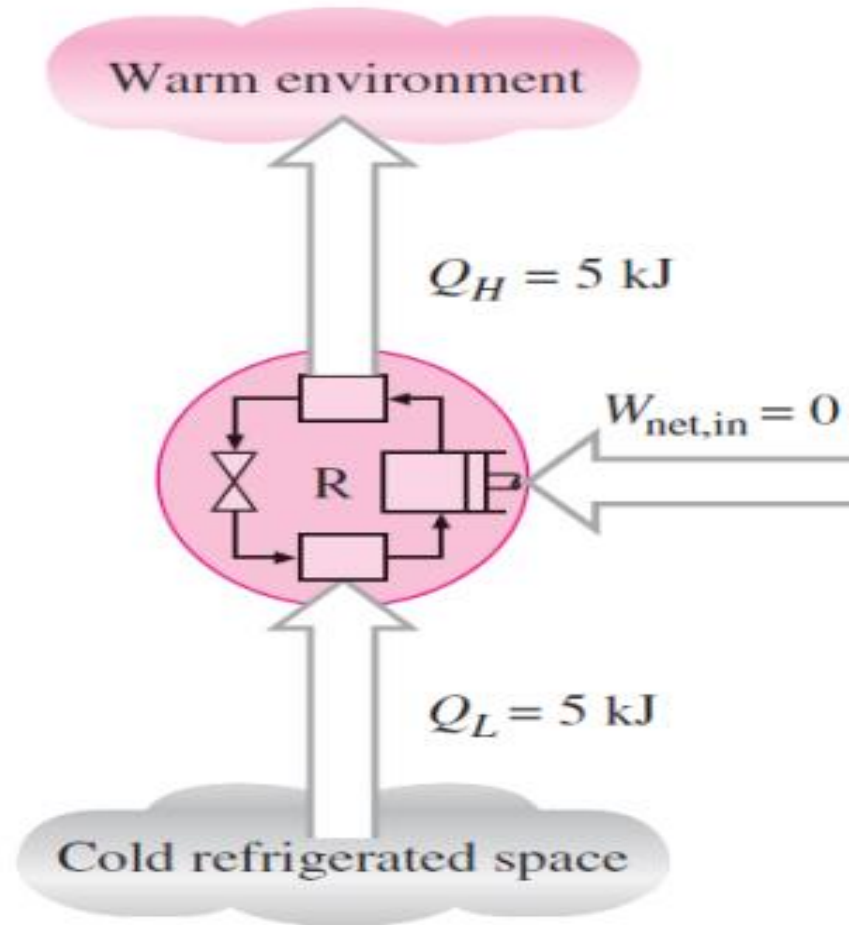


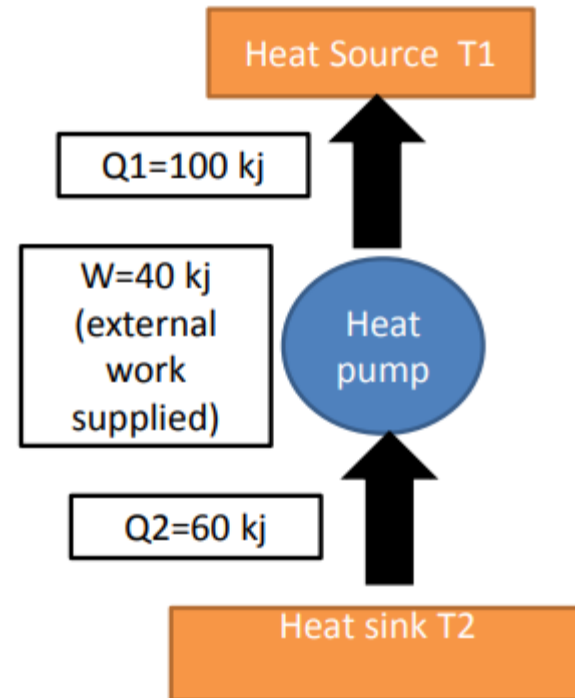
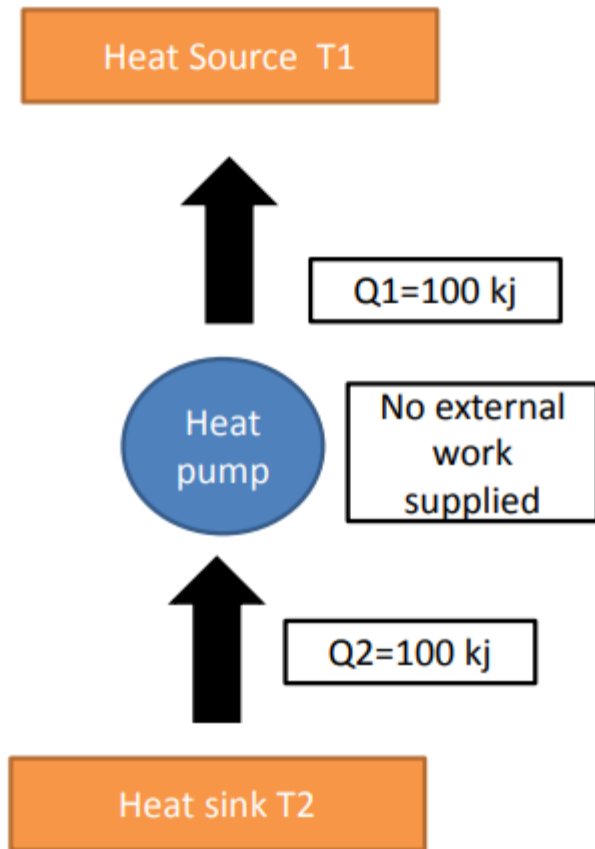
2.Clausius Statement

It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a lower-temperature body to a higher-temperature body

- Both the Kelvin–Planck and the Clausius statements of the second law are negative statements, and a negative statement cannot be proved. Like any other physical law, the second law of thermodynamics is based on experimental observations

Violation of Clausius statement





Equivalence of Kelvin-Planck and Clausius Statements

- Violation of the Clausius statement leads to the violation of the Kelvin–Planck statement

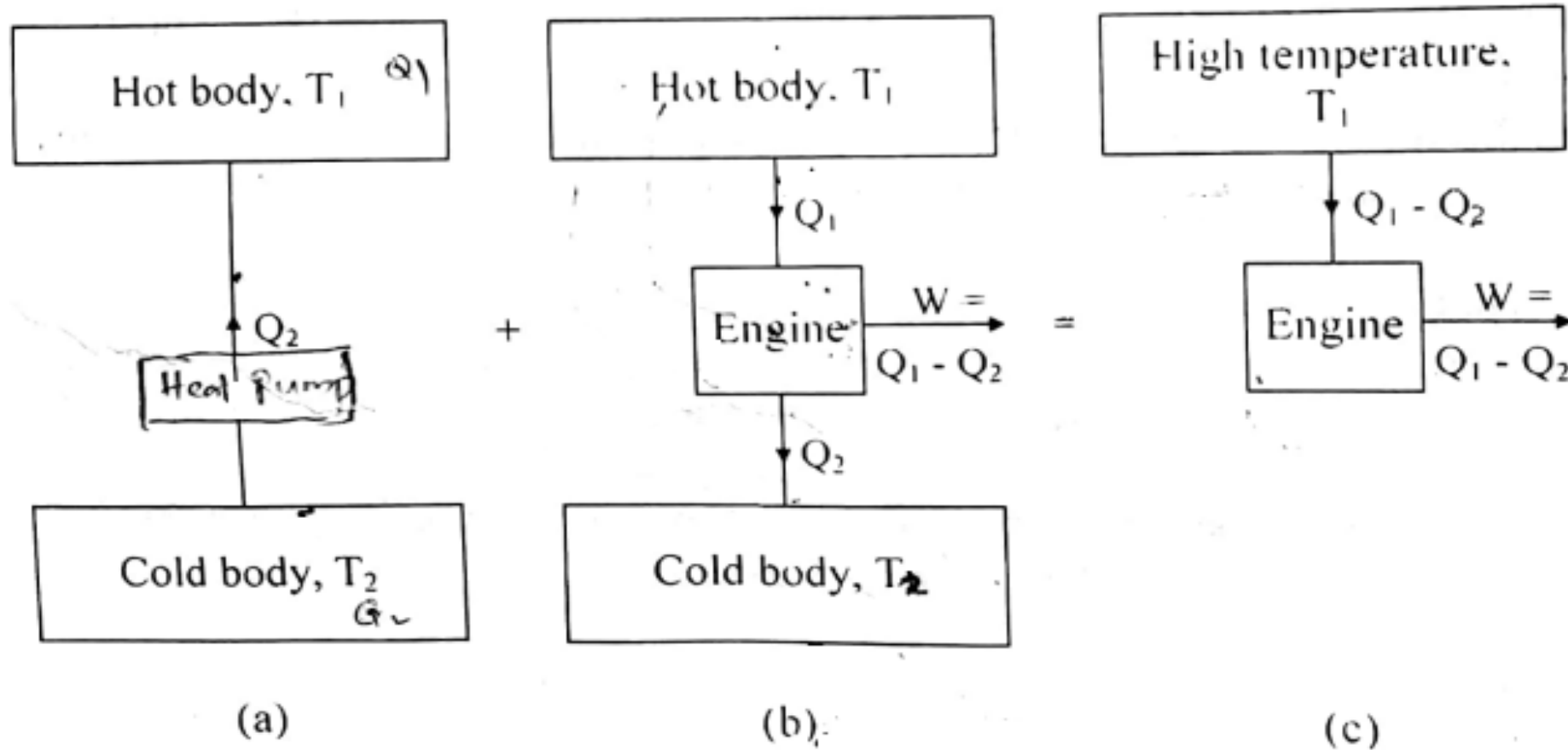


Fig. Violation of Clausius statement

Equivalence of Kelvin-Planck and Clausius Statements

- Violation of the Kelvin–Planck statement leads to the violation of the Clausius statement

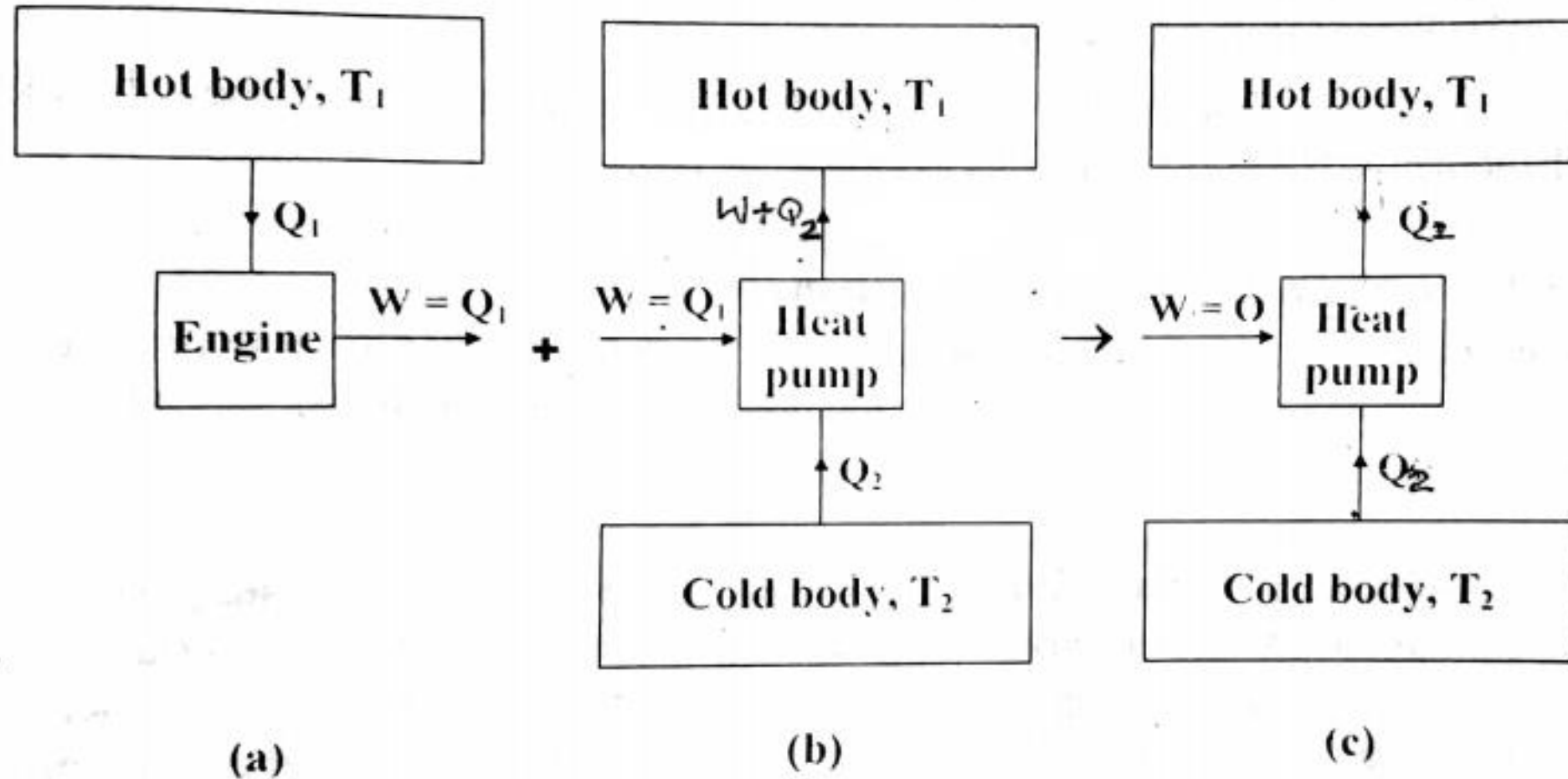


Fig. *Violation of Kelvin - Planck Statement*

COMPARISON

Kelvin-Plank Statement

- It is applied to 'Heat Engine'.
- It is negative statement.
- It is based on experimental observations and no mathematical proof.

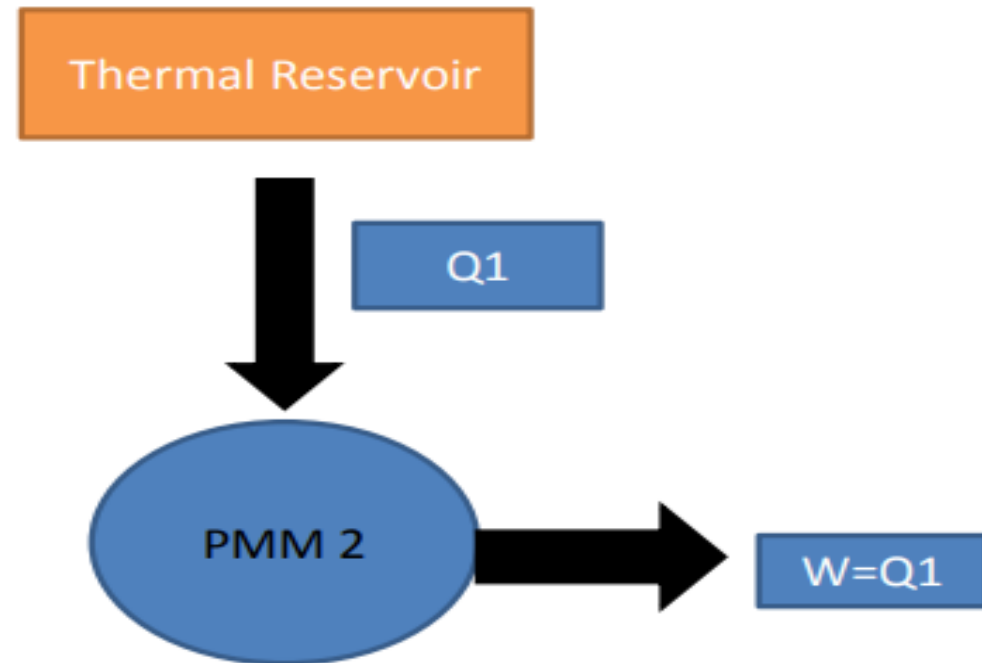
Clausius Statement

- It is applied to 'Heat Pump' and 'Refrigeration'.
- It is also negative statement.
- It is based on experimental observations and no mathematical proof.

Perpetual Motion Machine

PMM 2

- If the engine exchange heat **only single thermal reservoir**, in which heat is supplied is completely converted into an equivalent amount of work and its **efficiency becomes 100 %**.
- This kind of machine is known as "**PMM 2**".



ENTROPY

- Entropy can be viewed as a *measure of molecular disorder, or molecular randomness*. As a system becomes more disordered, the positions of the molecules become less predictable and the entropy increases. Thus, it is not surprising that the entropy of a substance is lowest in the solid phase and highest in the gas phase
- Entropy is the thermodynamic property of a working substance that increases with the increase of temperature and decreases with decrease of temperature
- Entropy cannot be measured directly, but the change of entropy during any process can be calculated
- It is a point function
- $dS = \delta Q/T$